Lecture 22

- Transposes
- Orthogonal vectors + subspaces
- Null space $\perp$ row space
- $N(A^TA) = N(A)$

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \rightsquigarrow A^T = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix}$$

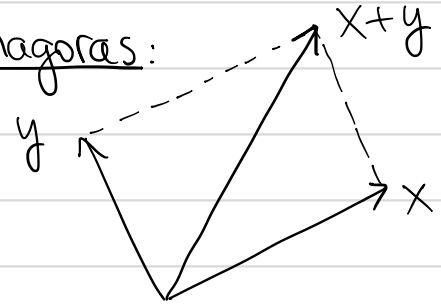
Remarks:

1) $(AB)^T = B^TA^T$
2) $(A^T)^{-1} = (A^{-1})^T$ when A^{-1} exists

   $A^T \cdot (A^{-1})^T = (A^{-1}A)^T = I^T = I$
3) $\underbrace{y^Tx}_{\text{inner product (= dot)}} = \begin{bmatrix} y_1 & y_2 & y_3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = y_1x_1 + y_2x_2 + y_3x_3 = \underline{\underline{x^Ty}}.$
4) $\underbrace{x^Tx}_{} = x_1^2 + \cdots + x_n^2 \geq 0$

   $\|x\|^2 =$ length squared.

Pythagoras:



$\|x\|^2 + \|y\|^2 = \|x+y\|^2$ if $\underbrace{x^Ty = 0}_{}$

x and y are <u>orthogonal</u> ($x \perp y$)

Example:

$$x = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \qquad y = \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} \qquad x+y = \begin{bmatrix} 3 \\ 1 \\ 3 \end{bmatrix}$$

$\|x\|^2 = 14 \qquad \|y\|^2 = 5 \qquad \|x+y\|^2 = 19$ ✓

$x^Ty = 0$

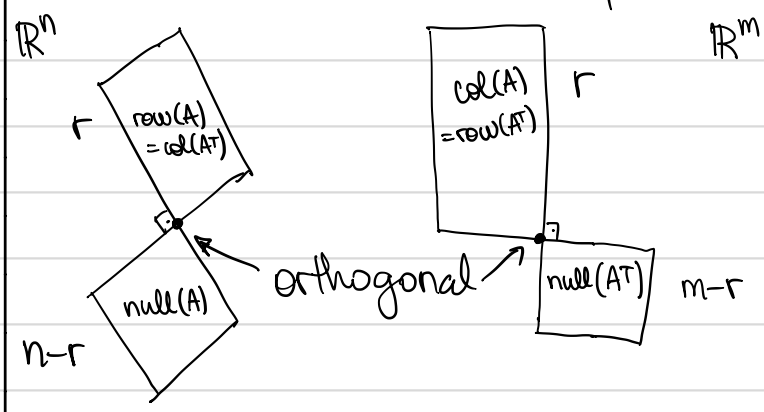
Why:

$$\begin{aligned} \|x+y\|^2 &= (x+y)^T(x+y) \\ &= x^Tx + x^Ty + y^Tx + y^Ty \\ &= \|x\|^2 + \|y\|^2 + 2 \cdot \cancel{x^Ty}^{\,0} \end{aligned}$$

$x \perp y$

A is m by n.

$\rightarrow$ Four (fundamental) subspaces



"Subspace S is orthogonal to subspace T" means: every vector in S is orthogonal to every vector in T.

null(A) is orthogonal to row(A):

$$Ax = 0 : \begin{bmatrix} \cancel{\text{row 1 of } A} \\ \text{row } 2 \\ \vdots \\ \text{row } n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ \vdots \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ \vdots \end{bmatrix}$$

$(\text{row } 1)^T x = 0 \quad (\text{row } 2)^T x = 0 \quad \dots$

$\Rightarrow (c_1(\text{row } 1) + c_2(\text{row } 2) + \dots)\, x = 0$

null(A) and row(A) are orthogonal <u>complements</u>:

null(A) contains every vector perpendicular to row(A).

<u>Coming</u>: "Solve" $Ax = b$ when there is no solution.

$\downarrow$

$A^TA\hat{x} = A^Tb.$

$$\begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

Find $A\hat{x}$ "closest" to b.

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1) $(A^TA)^T = A^TA^{TT} = A^TA$

$(n \times m)(m \times n) = \underline{n \times n}$ <u>symmetric</u>.

2) $\text{rank}(A^TA) = \text{rank}(A)$.

<u>Ex</u>: • $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$ $r = 2$

• $\begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 1 & 2 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 6 & 12 \end{bmatrix}$ $r = 1$

3) A^TA is invertible if A has independent columns.

<u>Why</u>:

Suppose $A^TAx = 0$. To prove: $x = 0$.

Idea: $x^TA^TAx = 0$

$\quad \| $

$(Ax)^T(Ax) = \|Ax\|^2 \Rightarrow Ax = 0$

$\Rightarrow x = 0$ since columns are indep. ∎